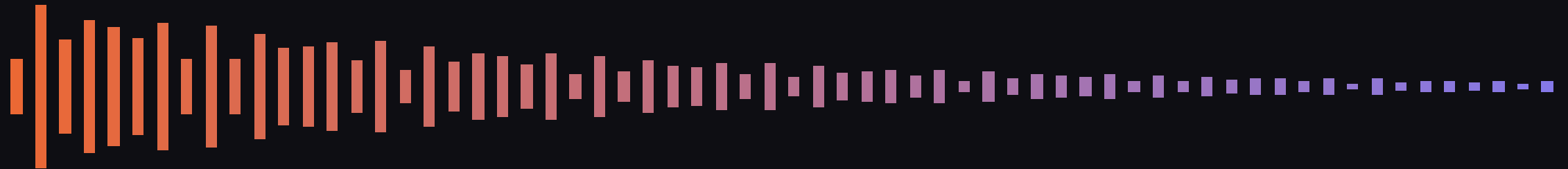


Is Music Becoming Flatter?

Is flatter music becoming more popular over time?



Two datasets, one question

● Chordonomicon — Harmony

Hugging Face · ailsntua/Chordonomicon (Kantarelis et al., 2024)
239,647 rows · pop / rap / rock · key columns: sid, anno, genre_grp, prog4

sid	anno	genre_grp	prog4 (4-chord blocks)
1	2003	rock	I-V-V-IV I-V-IV-IV I-I-V-IV ...
2	2003	rock	bVI-i-bVI-i bVI-bVII-i-i ...
3	2011	pop	I-V-vi-IV I-V-vi-IV vi-IV-I-V ...
4	2018	rap	i-bVI-bIII-bVII i-bVI-bIII-bVII ...

● 550K Spotify Songs — Lyrics + Audio

Kaggle · serkantysz/550k-spotify-songs (CC BY-NC-SA 4.0)
453,030 rows · pop / rap / rock · key columns: id, year, genre_grp, guiraud, popularity

id	year	genre_grp	guiraud	popularity
0Prct5T...	2016	rock	10.20	NA
2ASl4wi...	2019	rap	5.80	29
5kQIPFB...	2007	pop	5.41	42
7hDVYcQ...	2021	pop	6.13	55

Preprocessing

- 10 raw genres → 3 classes; keyword mapping, priority rap > pop > rock
- Popularity = 0 treated as missing (27% of rows)
- Deduplicated by title + artist, most popular version kept
- Chords parsed to Roman numerals; prog4 = exact 4-chord blocks, rotations collapsed
- Lexical complexity = Guiraud = unique words / $\sqrt{(\text{total words})}$
- Harmonic complexity = Guiraud on 4-grams = unique prog4 / $\sqrt{(\text{total blocks})}$

Where both datasets meet

≈ 28,000

songs present in both datasets
with popularity, lyrics and chords · 2000–2023

- Joined via the Spotify track ID shared by the two sources
- Every row carries both complexity measures and a popularity score
- Used for: correlation between lexical and harmonic complexity
- Used for: the flatness → popularity regression, and the hits analysis

● The Merged Table

spotify_song_id	genre_grp	year	guiraud	harm	popularity
2ASl4wi...	rap	2019	5.80	1.21	29
0Prct5T...	rock	2016	10.20	1.84	47
5kQIPFB...	pop	2007	5.41	1.95	42
7hDVYcQ...	pop	2021	6.13	1.60	55

Four questions

The roadmap of the analysis

1

Is music becoming lexically poorer?

2

Is music becoming harmonically less complex?

3

Are songs of each genre becoming more similar to themselves over time?

4

Do simpler songs become more popular over time?

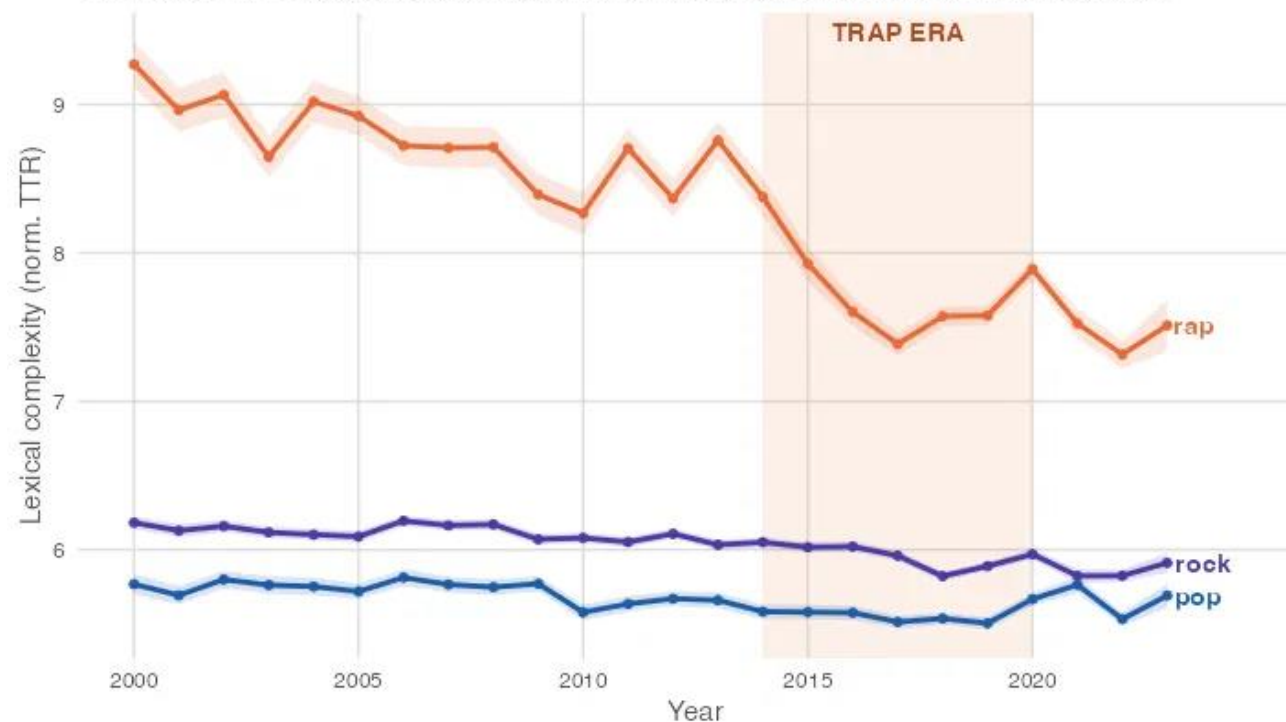
Is music becoming lexically poorer?

Normalized TTR (Guiraud) per song, yearly mean by genre · band = 95% CI

$$G = V(N) / \sqrt{N} \quad (V = \text{unique words}, N = \text{total words}) \quad \cdot \sqrt{N} \text{ corrects for song length}$$

Lexical complexity over time, by genre

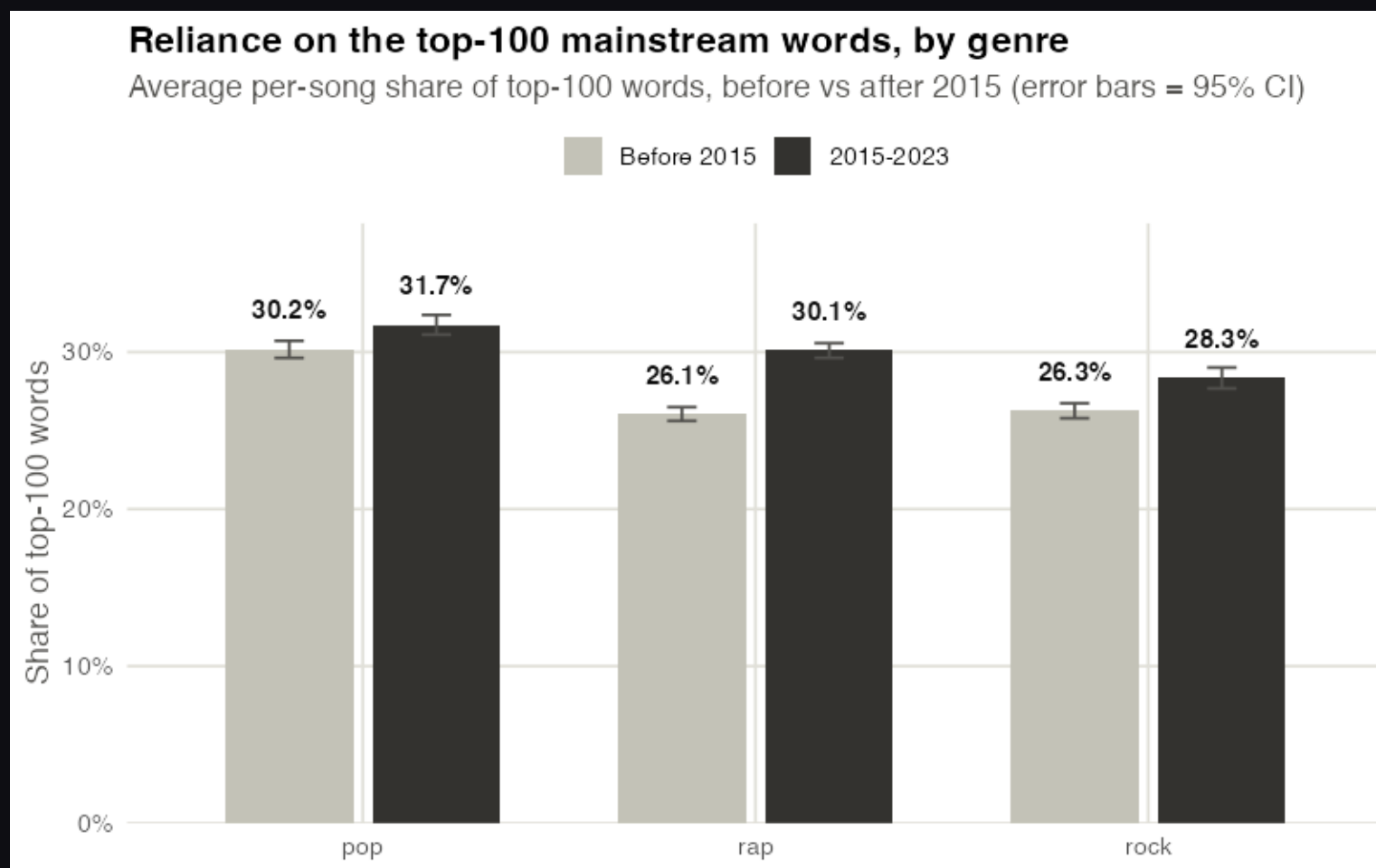
Normalized TTR = unique / sqrt(total) words (Guiraud); band = 95% CI of the yearly mean



Do songs always use the same words?

Share of a song's content words that are among the 100 most common overall

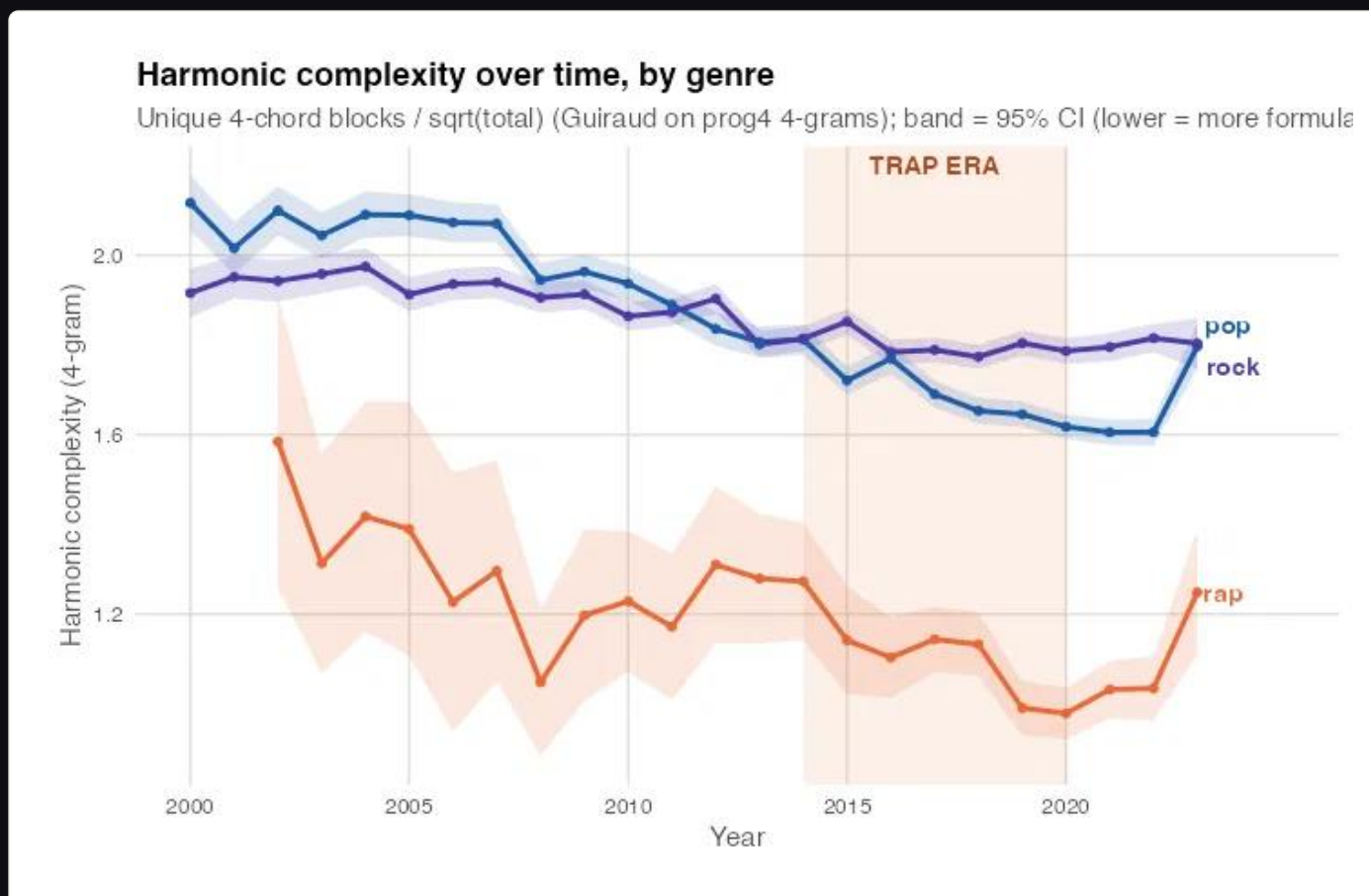
$\text{reliance} = |\text{song words} \in \text{Top100}| / |\text{song words}|$ · *Top-100 = most frequent content words (stop-words removed) · cutoff is arbitrary, complement to Guiraud*



Is music becoming harmonically less complex?

Guiraud on prog4 4-grams · lower = more formulaic · band = 95% CI

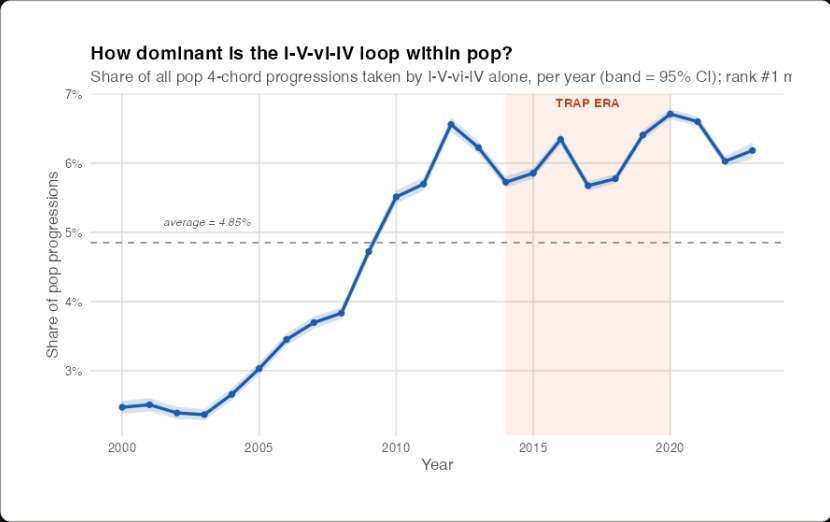
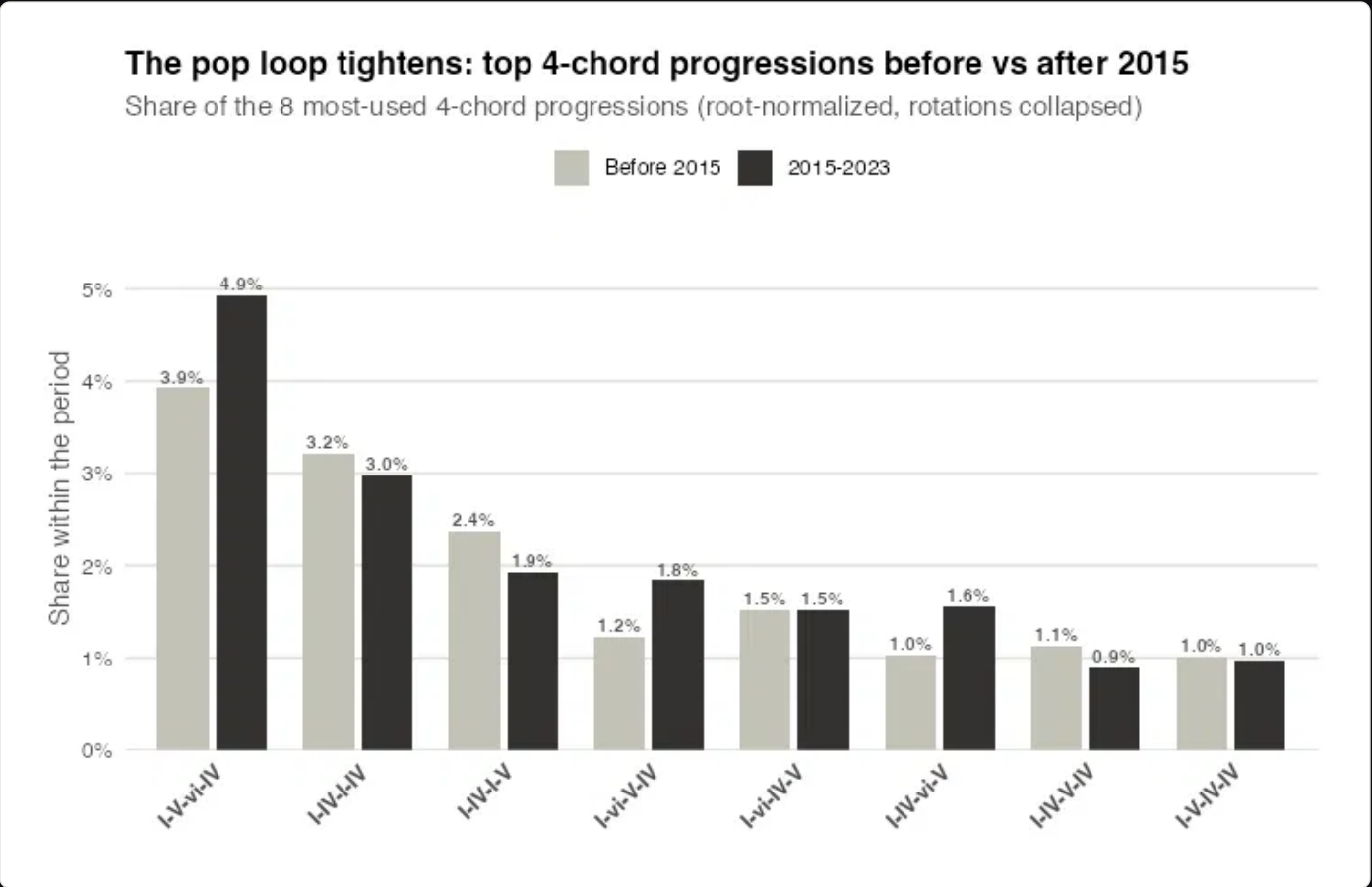
$H_g = U(\text{prog4}) / \sqrt{N(\text{prog4})}$ · prog4 = 4-chord blocks, key-normalized, rotation-collapsed · symmetric to the lexical metric



In the post-algorithm era, are the same chords always used?

Share of the 8 most-used 4-chord progressions, before 2015 vs 2015–2023

I-V-vi-IV: #1 progression overall, share 2.5% → 6.2% (2000–2023) · the single loop that keeps gaining ground

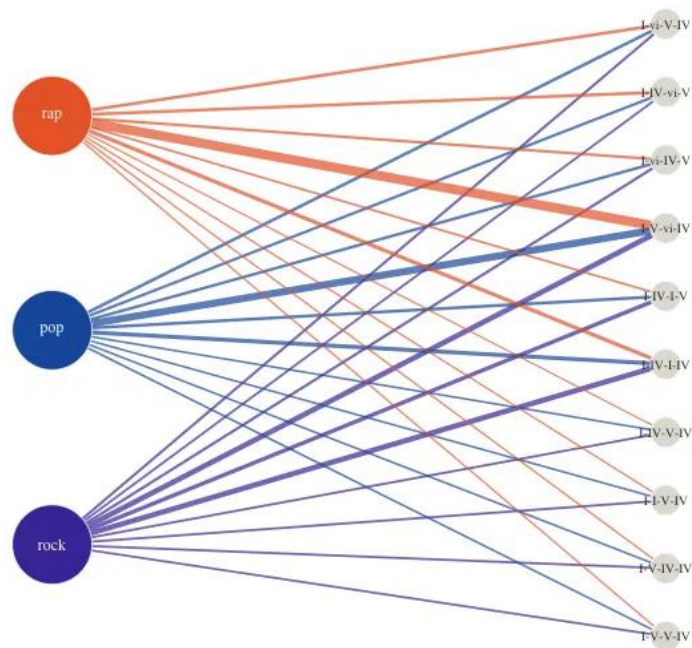


Which chords characterize each genre?

Bipartite network: genres $\leftrightarrow$ top 4-chord progressions · edge width = reliance

$\mathbf{A} = [\mathbf{a}_{ik}] \quad (3 \times k)$ $(\mathbf{A}\mathbf{A}^T)_{ij} = \sum_k \mathbf{a}_{ik} \cdot \mathbf{a}_{jk} \quad \cdot \quad a_{ik} = \text{share, so the projection is a dot product of profiles: shared harmonic vocabulary between genres } i, j$

Harmonic bipartite network: which 4-chord progressions each genre leans on



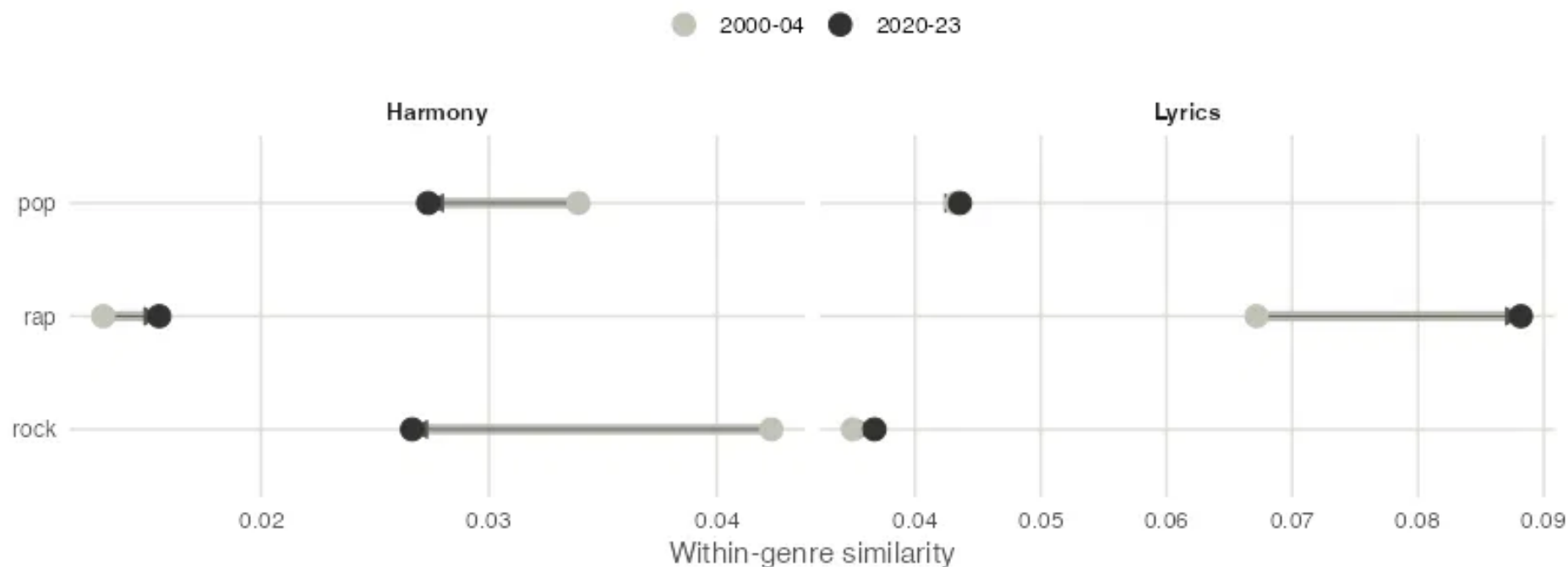
Are songs within a genre becoming more similar?

Average pairwise cosine similarity within a genre, 2000–04 vs 2020–23

$$\cos(i, j) = (\mathbf{x}_i \cdot \mathbf{x}_j) / (\|\mathbf{x}_i\| \|\mathbf{x}_j\|) \quad \cdot \quad \text{mean of the off-diagonal entries of } S = M \cdot M^T \text{ (row-normalized)}$$

Did genres grow more self-similar? Mostly no - only rap lyrics did

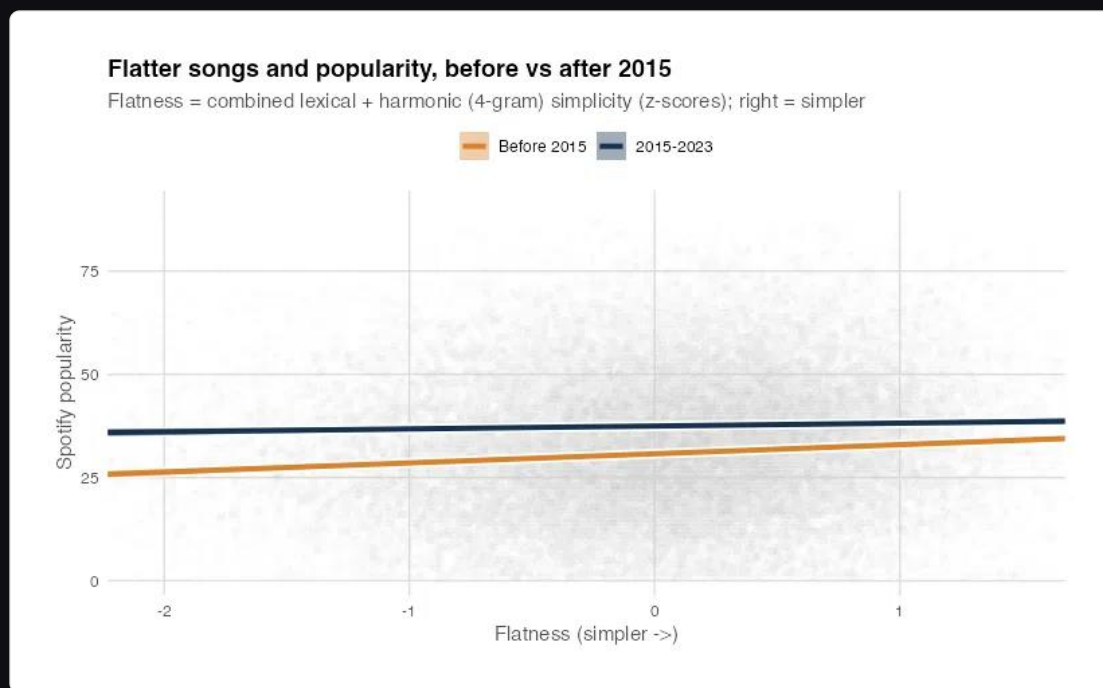
Average pairwise cosine similarity within a genre, 2000-04 vs 2020-23 (right = more self-similar)



Does a song's flatness influence its popularity?

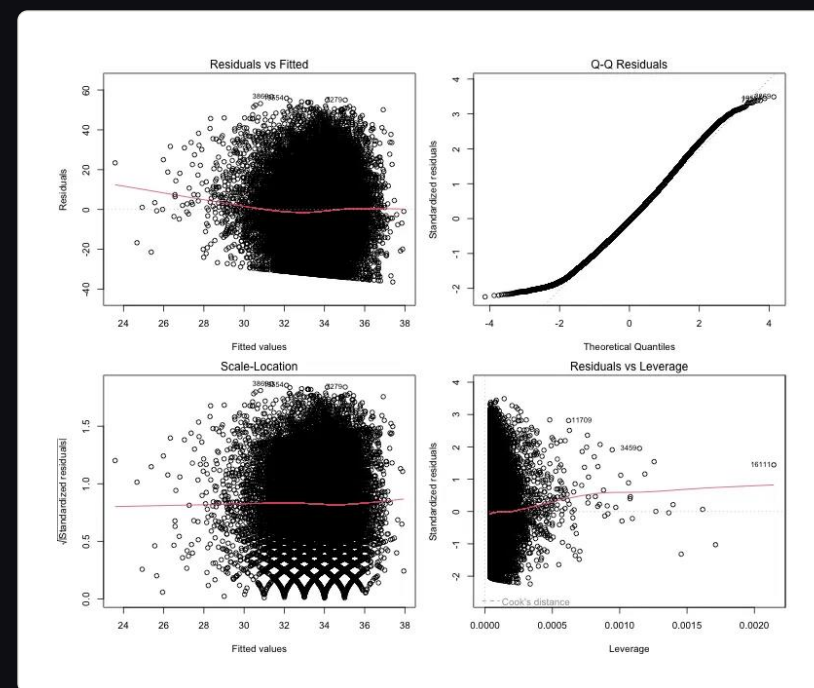
Model 1 — popularity ~ flatness (combined lexical + harmonic simplicity, z-scores), with residual diagnostics

$$\text{flatness} = -(z_{\text{lex}} + z_{\text{harm}}) / 2 \quad y = \beta_0 + \beta_1 X + \varepsilon \quad (\text{OLS}) \quad \cdot \text{higher flatness} = \text{simpler song}$$



Fit: Positive But Shallow

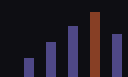
Flatter songs are somewhat more popular, but the slope is small and it flattens further after 2015 — the effect is real, not large.



Diagnostics: Assumptions Hold, Loosely

Residuals are centred and homoscedastic; Q-Q is heavy in the upper tail (popularity is bounded at 100) and no point exceeds Cook's distance.

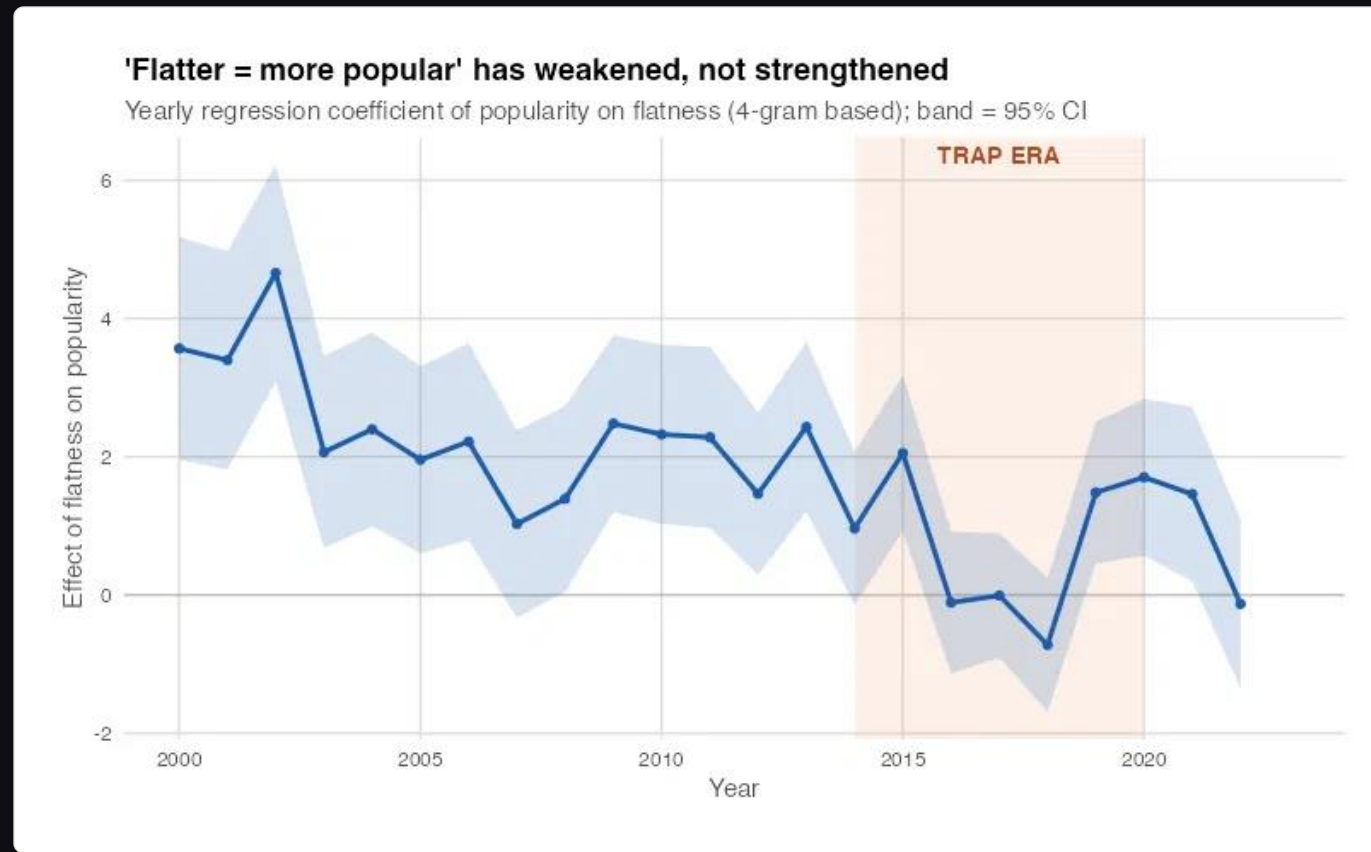
Assumptions: linearity · independence · homoscedasticity (Breusch-Pagan) · normality (Q-Q)



How much does flatness influence popularity — over time?

Yearly regression coefficient of popularity on flatness · band = 95% CI

per year: $\text{popularity} = \beta_0 + \beta_1 \cdot \text{flatness} + \varepsilon$ · ● filled = 95% CI excludes 0 ○ empty = not significant



Assumptions: linearity · independence · homoscedasticity (Breusch-Pagan) · normality (Q-Q)

What actually predicts a song's popularity?

Multiple regression: lexical vs harmonic simplicity, pooled vs genre-controlled

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 \text{rap} + \varepsilon$$

	Pooled (m1)	+ rap (m2)
lex_simpl	-0.32 **	+0.79 ***
harm_simpl	+2.11 ***	+1.33 ***
rap dummy	—	+18.55 ***
R ²	0.017	0.095

ANOVA (m1 vs m2): $F(1, 27837) = 2400.5$, $p < 2.2e-16$

VIF: $1.06 \cdot 1.03 \cdot 1.08$ — no multicollinearity

$n = 27,841$ · *** $p < 0.001$, ** $p < 0.01$

● Simpson's Paradox

Pooled, lexical simplicity looks harmful (-0.32). Add the rap control and the sign flips: +0.79.

The pooled coefficient was never about words. It was rap — lexically rich, and +18.55 points more popular.

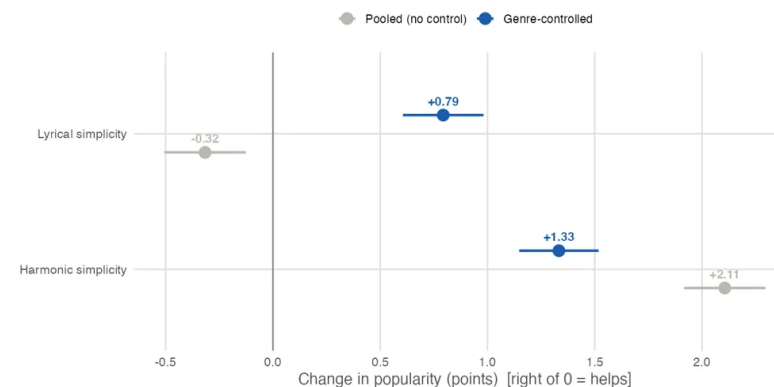
Even so, $R^2 = 0.095$. The song itself explains under 10% of popularity.

What Predicts Popularity — and How Has It Changed Over Time?

Multiple regression on lexical & harmonic simplicity (top) · yearly effect of flatness on popularity (bottom)

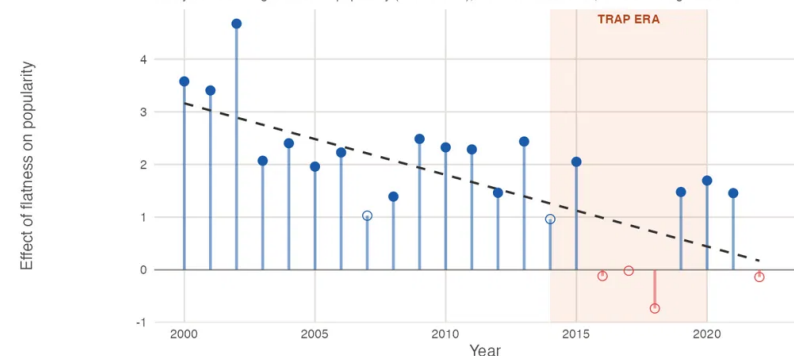
What predicts popularity?

Effect on popularity per +1 SD of simplicity (95% CI). Pooling flips the lyrical sign (Simpson's paradox).



How did flatness's effect on popularity change over time?

Yearly effect of song flatness on popularity (stem from 0); filled = CI excludes 0; dashed = weighted trend



Assumptions: linearity · independence · homoscedasticity (Breusch-Pagan) · normality (Q-Q)



The regression output

Model 1 (pooled) vs Model 2 (+ rap control) · nested ANOVA · variance inflation factors

● summary(m1) — pooled

```
lm(formula = popularity ~ lex_simpl + harm_simpl)
```

Coefficients:

	Estimate	Std.Error	t value	Pr(> t)	
(Intercept)	33.61711	0.09679	347.31	< 2e-16	***
lex_simpl	-0.31627	0.09700	-3.26	0.00111	**
harm_simpl	2.10578	0.09700	21.71	< 2e-16	***

Residual standard error: 16.15 on 27838 DF

Multiple R-squared: 0.01677 Adjusted: 0.01669

F-statistic: 237.3 on 2 and 27838 DF, p-value: < 2.2e-16

● summary(m2) — with the rap control

```
lm(formula = popularity ~ lex_simpl + harm_simpl + mainstream)
```

Coefficients:

	Estimate	Std.Error	t value	Pr(> t)	
(Intercept)	32.31760	0.09659	334.602	<2e-16	***
lex_simpl	0.79362	0.09579	8.285	<2e-16	***
harm_simpl	1.33287	0.09440	14.119	<2e-16	***
mainstreamrap	18.55365	0.37868	48.995	<2e-16	***

Residual standard error: 15.50 on 27837 DF

Multiple R-squared: 0.09482 Adjusted: 0.09473

F-statistic: 972 on 3 and 27837 DF, p-value: < 2.2e-16

● anova(m1, m2) · vif(m2)

```
anova(m1, m2)      Analysis of Variance Table
```

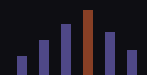
Model 1: popularity ~ lex_simpl + harm_simpl

Model 2: popularity ~ lex_simpl + harm_simpl + mainstream

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	27838	7261096				
2	27837	6684647	1	576449	2400.5	< 2.2e-16 ***

vif(m2)	lex_simpl	harm_simpl	mainstream
	1.063797	1.033155	1.082920

$F = 2400.5$: the rap control is a massive improvement. $VIF \approx 1$: the two complexity measures are orthogonal. $R^2 = 0.095$ all the same.



What the data show — and don't

● What the data show

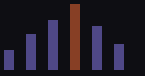
- Music is getting flatter: harmonically for every genre, lexically for rap
- The pop loop tightens: I-V-vi-IV keeps gaining share after 2015
- Flatness does predict popularity — but weakly
- And the effect is fading over time, not growing

● What the data don't show

- No causality: 2015 is a descriptive cutoff, not an "algorithm" variable
- No proof that flatness drives streaming-era success
- Genres did not converge on themselves — only rap lyrics did
- Content explains little: $R^2 = 0.095$ even with the genre control

● Limits

- English lyrics only
- Spotify year reflects album reissue dates
- Chordonomicon uses "rap" as a proxy for trap
- Correlational, not causal, throughout
- Independence violated: songs are not i.i.d. (same artist / same year)



Thank You for Your Attention

Is Music Becoming Flatter? — Questions welcome

